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Dept

AI - 4-1

Quiz

Algorithm

Date

24/04/26

Huffman Coding :-

BD KFGDKGF DKB

B = 20

D = 15

F = 10

G = 5

K = 40

Sorting:-

G = 5

F = 10

D = 15

B = 20

K = 40

DATE: ___/___/___

DAY: ___/___/___

Construction

$$G(5) + F(10) \rightarrow 15$$

Now

$$15(GF), 15(D), 20(B), 40(K)$$

$$15 + 5 = 30$$

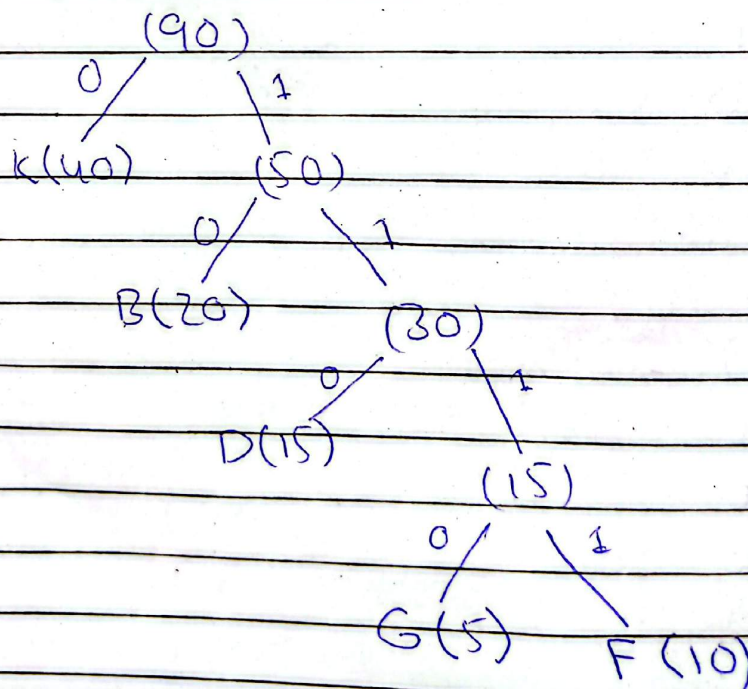
$$20(B), 30, 40(K)$$

$$20 + 30 = 50$$

$$40(K), 50$$

$$40 + 50 = 90$$

Tree :-



DATE: ___/___/___

DAY: ___/___/___

So the codes are

K $\rightarrow$ 0

B $\rightarrow$ 10

D $\rightarrow$ 110

G $\rightarrow$ 1110

F $\rightarrow$ 1111

Encoding:-

~~BDF~~

B D K F G D K G F D K B

10 110 0 111 1110 110 0 1110
1111 110 0 10

101100111111011001110111110010

Decoded:-

10 = B

110 = D

0 = K

B D K F G D K G F D K B